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Computer Science
Department

End of Studies Project Report

For the obtention of a National Engineering Diploma

Track: Software Engineering and Data Engineering

Title:

PLAN GENIE AI

Intelligent Task & Event Management Platform

Powered by Advanced NLP and Speech Recognition

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Realised within **SFM**



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Dedication

*To my family and mentors,
whose support made this possible.*

— *Yassine Jedidi*

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Introduction

In today's busy world, managing tasks and events effectively is very important for both personal and work productivity. Traditional ways of managing tasks and events are often slow, make mistakes, and don't adapt to what users prefer.

This project introduces **Plan Genie AI**, a smart task and event management platform that uses artificial intelligence to change how people and organizations plan, organize, and track their activities. The system uses Natural Language Processing (NLP), speech recognition, and data analysis to provide a smooth, intelligent, and easy-to-use experience.

Main Goals:

- Build an AI platform that can understand and process natural language for creating tasks and planning events
- Add speech-to-text features to allow voice-based task and event management
- Provide performance analytics and visualization tools
- Ensure data security and privacy protection
- Make a responsive application that works on different devices

The project uses modern technologies including React.js for the frontend, Node.js and Python FastAPI for backend services, PostgreSQL for data storage, and Hugging Face Transformers for advanced NLP features. The system can process multiple languages, extract information automatically, and provide detailed performance analytics through interactive charts.

This report shows the complete development process of Plan Genie AI, from initial idea and requirements through design, building, testing, and deployment. It demonstrates how artificial intelligence can be used to solve real productivity problems while maintaining good user experience and data security.

1 Preliminary Study

Introduction

The modern workplace and personal life demand efficient management of tasks and events. Traditional task management systems often fall short due to their rigid interfaces, limited automation capabilities, and lack of intelligent features. This chapter provides a comprehensive preliminary study of existing task and event management solutions, analyzing their strengths, limitations, and identifying market gaps that Plan Genie AI aims to address. Through this analysis, we establish the foundation for understanding current market limitations and opportunities for innovation in intelligent task and event management.

1.1 Host Organization: SFM Technologies

1.1.1 General Overview

Founded in 1995, **SFM Technologies** is a Tunisian consulting and engineering firm specializing in telecommunications, ICT, and digital transformation services. The company is part of the larger SFM Group, which includes subsidiaries such as SFM International, SFM Telecom, and SFM Cameroun. SFM operates in Africa, Europe, Asia, Oceania, and the United States.

1.1.2 Core Competencies

SFM Technologies provides a wide array of services in the following areas:

- **Telecommunications and Regulatory Support:** Benchmarking, audits, QoS/QoE analysis, and network planning for regulators and operators.
- **Strategic Engineering and Advisory:** Consulting on network convergence, privatization, spectrum regulation, and technology migration (e.g., analog to digital TV).
- **Training and Capacity Building:** Technical and theoretical training sessions in ICT and telecoms for institutions and companies.
- **R&D and Software Development:** Through its R&D branch, SFM Lab, the company creates custom software for fraud detection, tariff simulation, AI analytics, and network performance monitoring.

1.1.3 Digital Tools and Platforms

SFM Technologies develops and maintains several proprietary platforms:

- Robotic Process Automation (e.g., COSAP, Ticketeazy, IT&M)
- Cybersecurity Solutions (e.g., WiFi portal, PKI, SOC)
- AI and Big Data platforms for telecom, banking, and government sectors
- Modular software solutions tailored to specific client needs

1.1.4 Clientele and Global Reach

With a portfolio of over 30 clients, including national regulators, ministries, and international organizations such as the World Bank and African Development Bank, SFM Technologies has established itself as a trusted partner across several continents.

1.1.5 Summary Table

Founded	1995
Legal Form	SARL (R.C. 1028016M)
Headquarters	Tunis, Tunisia
Core Areas	Telecom regulation, digital transformation, ICT consulting and training
R&D Division	SFM Lab
Operational Presence	Africa, Europe, Asia, Oceania, USA
Main Offerings	Consulting, audits, custom platforms, cybersecurity, AI solutions
Clients	Telecom operators, regulators, ministries, international donors

Table 1: Summary of SFM Technologies’ Key Information

1.2 Project Context and Problem Statement

1.2.1 Project Context

The Plan Genie AI project emerged from a real-world need identified during the internship at SFM Technologies. The company, being a leading ICT consulting firm with operations across multiple continents, frequently deals with complex project management scenarios involving diverse teams, multilingual environments, and tight deadlines using their proprietary platform named COSAP.

Recognizing the transformative potential of artificial intelligence, SFM Technologies sought to introduce AI-powered capabilities into their task and event management ecosystem. This strategic decision to leverage advanced AI technologies for enhanced productivity and intelligent automation ultimately led to the foundation of Plan Genie AI as a comprehensive, AI-driven task and event management platform.

Project Organization and Planning To ensure successful delivery of the Plan Genie AI platform, the project was structured into well-defined phases, each with specific objectives and deliverables. This systematic approach helps track progress, manage resources effectively, and ensure quality throughout the development process.

Project Phases

- 1. **Phase 1: Conception** - Requirements analysis, system design, and architecture planning
- 2. **Phase 2: Development** - Implementation of core features and AI integration
- 3. **Phase 3: Tests and Validation** - Quality assurance and performance optimization

Project Deliverables

The project produces several key deliverables that demonstrate the completion of objectives and provide value to both the development team and end users. These deliverables serve as milestones and evidence of project progress.

Deliverables

- Functional and technical documentation
- Validated mockups and wireframes
- Complete and structured source code
- Functional application (web and mobile versions)
- Test reports and validations

Project Constraints and Requirements Every project operates within certain boundaries and must meet specific requirements to be considered successful. Understanding these constraints and requirements is crucial for proper planning and execution.

Constraints and Requirements

- **Multi-platform Compatibility:** Support for web, mobile, and desktop platforms
- **Performance and Scalability:** Efficient handling of large datasets and concurrent users
- **Data Security:** Implementation of robust security measures and data protection
- **Regulatory Compliance:** Adherence to relevant data protection and privacy regulations
- **Accessibility and Ergonomics:** User-friendly interface with accessibility features

Project Timeline and Schedule

Effective project management requires realistic time planning and scheduling. SFM Technologies defined a well planned and organized timeline. This timeline provided a roadmap for project execution and helps identify critical path activities that could impact overall project delivery.

Estimated Timeline

Phase	Estimated Duration
Conception	4 weeks
Development	16 weeks
Tests and Validation	4 weeks
Total	24 weeks (6 months)

Table 2: Project Timeline Overview

1.2.2 Problem Statement

Primary Problem

Modern teams and individuals capture work as unstructured notes, chats, and voice memos. Converting these inputs into well-formed tasks and calendar events with priorities, deadlines, participants, and context remains largely manual. Existing tools provide partial solutions (basic date parsing, simple reminders, or isolated analytics) but lack an end-to-end, multilingual pipeline that: (1) ingests free-form text and speech, (2) extracts reliable entities, (3) generates tasks/events, (4) prioritizes them intelligently, and (5) surfaces actionable analytics—while meeting privacy, scalability, and usability requirements. Within SFM Technologies’ context (e.g., projects managed in COSAP), this gap leads to friction, inconsistent planning quality, and missed opportunities for data-driven productivity improvements.

Research Question

How can we design and implement an AI-driven platform that transforms free-form text and voice into structured, prioritized tasks and events with multilingual NER, intelligent scheduling/prioritization, and actionable analytics—integrated into a secure, cross-platform experience suitable for enterprise use cases like SFM Technologies—while respecting performance, privacy, and accessibility constraints?

Scope and Boundaries

- **In-scope:** Multilingual NLP for entity extraction, voice-to-text transcription, automatic task/event generation, AI-assisted prioritization, interactive calendar and dashboards, user authentication and data persistence, and responsive web/mobile UI.
- **Out-of-scope (for this iteration):** Full project portfolio management, advanced multi-user collaboration (role workflows beyond sharing and sync), deep third-party ecosystem integrations (beyond API-ready design), and offline-first with conflict-free replication.

Constraints and Non-Functional Requirements

- **Privacy and Security:** Protect user data at rest and in transit; align with GDPR principles; integrate with secure authentication and role-based access.
- **Performance:** Low-latency text/voice processing suitable for interactive UX; scalable APIs for concurrent users.
- **Multilinguality and Accessibility:** Support multiple languages across NLP and UI; ensure accessible interaction patterns.
- **Maintainability:** Modular services (NLP via FastAPI/Hugging Face, prioritization via Gemini) with clear contracts for evolution and testing.

Success Criteria (Measurable)

- **Entity Extraction Quality:** Accurate identification of dates, deadlines, priorities, participants, and contexts in supported languages, evidenced by internal validation on representative datasets.
- **Speech-to-Text Utility:** Transcriptions that are sufficiently accurate for downstream task/event generation in typical meeting/field conditions.
- **Prioritization Usefulness:** AI prioritization that aligns with expert-curated baselines and improves user-perceived planning quality (measured via task reordering acceptance and completion timeliness).
- **User Experience:** Responsive UI with intuitive flows; reduced manual steps from input to organized plan compared to baseline tools.
- **Operational Performance:** API endpoints for NLP, transcription, and prioritization respond within interactive thresholds under expected load.

Specific Challenges

- **Unstructured Input Understanding:** Robust NER over noisy, multilingual text.

- **Voice Integration:** Reliable speech-to-text that preserves task-critical details (dates, names, actions).
- **Intelligent Prioritization:** Context-aware ordering that reflects deadlines, dependencies, effort, and user preferences.
- **Analytics that Drive Action:** Visualizations that help users improve planning and execution, not just report activity.
- **Enterprise Readiness:** Security, compliance, and performance suitable for environments like SFM Technologies.

Impact Assessment

Within SFM Technologies’ workflows, streamlining the path from raw inputs (notes, voice) to structured, prioritized plans is expected to reduce planning time, increase adoption of the planning toolset, and improve on-time completion. In addition, multilingual capabilities address global operations, and integrated analytics support continuous improvement in individual and team productivity.

1.3 Study and Critique of Existing Solutions

To comprehensively understand the current landscape, we analyzed leading task and event management solutions across different categories: traditional task managers, calendar applications, project management tools, and AI-powered productivity platforms.

1.3.1 Traditional Task Management Solutions

Solution	Strengths	Limitations
Todoist	Clean interface, cross-platform availability, natural language date parsing	Limited AI features
Microsoft To Do	Integration with Microsoft ecosystem, simple interface	No AI features
Any.do	Voice input capability, calendar integration	Limited NLP

Table 3: Analysis of Traditional Task Management Solutions

1.3.2 Calendar and Event Management Solutions

Solution	Strengths	Limitations
Google Calendar	Excellent event management, natural language processing for dates	Limited task management
Outlook Calendar	Enterprise integration, meeting scheduling	Complex interface

Table 4: Analysis of Calendar and Event Management Solutions

1.3.3 AI-Powered Productivity Platforms

Solution	Strengths	Limitations
Notion AI	Advanced AI writing assistance, flexible workspace	Not specifically designed for task management
ClickUp AI	AI-powered task generation, project management features	Complex interface, expensive for small teams

Table 5: Analysis of AI-Powered Productivity Platforms

1.4 Proposed Solution and Objectives

1.4.1 Solution Overview

Plan Genie AI addresses the identified market gaps by providing an intelligent, AI-powered task and event management platform that combines advanced Natural Language Processing, speech recognition, and data analytics to deliver a seamless user experience.

1.4.2 Core Objectives

Primary Objectives:

1. Develop an intelligent task and event management system with advanced NLP capabilities
2. Implement multilingual support for global accessibility
3. Integrate speech-to-text functionality for voice-based interaction
4. Provide intelligent task prioritization using AI algorithms
5. Deliver comprehensive analytics and performance tracking
6. Ensure data security and privacy compliance

Secondary Objectives:

1. Create a responsive, cross-platform application
2. Implement real-time synchronization and collaboration features
3. Provide customizable notification and reminder systems
4. Develop an intuitive user interface with modern design principles

1.4.3 Key Differentiators

Advanced NLP Integration The platform leverages state-of-the-art Natural Language Processing models, including custom-trained Named Entity Recognition (NER) models, to automatically extract and categorize information from natural language inputs.

Multilingual Capabilities Unlike most existing solutions, Plan Genie AI supports multiple languages, making it accessible to a global user base and suitable for international organizations.

Intelligent Prioritization The system uses Google's Gemini AI to intelligently prioritize tasks based on context, deadlines, user preferences, and historical patterns.

Comprehensive Analytics Advanced data visualization and analytics provide users with insights into their productivity patterns, helping them optimize their workflow and time management.

1.4.4 Comparative Analysis

Feature	Todoist	Google Calendar	Notion AI	ClickUp AI	Plan Genie AI
Natural Language Processing	Basic	Limited	Advanced	Moderate	Advanced
Multilingual Support	Limited	Yes	Limited	Limited	Yes
Voice Input	No	No	No	No	Yes
AI-Powered Prioritization	No	No	No	Yes	Yes
Named Entity Recognition	No	No	No	Limited	Yes
Advanced Analytics	Basic	Basic	Moderate	Advanced	Advanced
Cross-Platform	Yes	Yes	Yes	Yes	Yes
Real-time Sync	Yes	Yes	Yes	Yes	Yes
Privacy Compliance	Basic	Advanced	Moderate	Advanced	Advanced

Table 6: Feature Comparison: Plan Genie AI vs. Existing Solutions

1.5 State of the Art on Named Entity Recognition Solutions

1.5.1 Introduction to Named Entity Recognition

Named Entity Recognition (NER) is a fundamental component of Natural Language Processing that involves identifying and classifying named entities in text into predefined categories such as person names, organizations, locations, dates, times, and quantities. In the context of task and event management, NER plays a crucial role in automatically extracting relevant information from natural language inputs.

1.5.2 Current NER Technologies

Traditional Rule-Based Approaches Early NER systems relied on handcrafted rules and dictionaries to identify entities. While effective for specific domains, these approaches lack flexibility and require extensive manual maintenance.

Statistical Methods Machine learning approaches using statistical models such as Hidden Markov Models (HMM) and Conditional Random Fields (CRF) improved NER performance but still required extensive feature engineering.

Deep Learning Revolution The introduction of deep learning models, particularly transformer-based architectures, has revolutionized NER capabilities:

- **BERT-based Models:** Bidirectional Encoder Representations from Transformers provide contextual understanding of text
- **RoBERTa:** Optimized BERT variant with improved training methodology
- **CamemBERT:** French-optimized BERT model for French language processing
- **XLM-RoBERTa:** Multilingual model supporting 100+ languages